S7 Formulas of secondary steps moved from the manuscript

To keep the manuscript within the page limit, the formulas below were moved from the method section. They restate definitions or standard expressions; every variable they introduce is also defined in the text of the manuscript, and the key steps of TiDAL-Net remain in the manuscript.

Normalized event time and input mask (Section III-A, after Eq. (1)). The onboard event epoch is normalized to mission elapsed time in nominal intervals, and the input mask marks valid observations on the grid:

$$\tilde{t}_{i,q} = \frac{a_{i,q} - a_{i,0}}{\Delta^{\mathrm{nom}}}, \qquad m^{\mathrm{in}}_{i,k} \in \{0, 1\}. \tag{S2}$$

Sources of irregularity (Section III-A, after Eq. (4)). Event intervals are non-uniform, the ground receiving time adds a path-dependent delay and a residual uncertainty to the event epoch, and the input masks of channels in the same subsystem are correlated:

$$\tilde{t}_{i,q} - \tilde{t}_{i,q-1} \neq \mathrm{const}, \tag{S3}$$

$$t^{\mathrm{rec}}_{i,q} = a_{i,q} + \Lambda_{\pi_{n(i)}}\left(a_{i,q}\right) + \eta^{\mathrm{rec}}_{i,q}, \tag{S4}$$

$$\mathrm{Cov}\left(m^{\mathrm{in}}_{i,k}, m^{\mathrm{in}}_{j,k}\right) \neq 0, \qquad i, j \in \mathcal{I}_{\mathrm{sub}}, \tag{S5}$$

where Λ_{π_n} is the delay along the communication path π_n of satellite n and $\eta^{\mathrm{rec}}_{i,q}$ is the residual receiving-time uncertainty. The onboard event time is unchanged.

Integral form of Bi-NSDE (Section III-B, after Eq. (7)). The latent trajectory of Eqs. (6) and (7) is the Itô process

$$\mathbf{h}^{(\mathrm{D})}_{i,k}(s) = \mathbf{h}^{(\mathrm{D})}_{i,k}(0) + \int_0^s f^{(\mathrm{D})}_\theta\left(\mathbf{h}^{(\mathrm{D})}_{i,k}(u), \mathbf{c}^{(\mathrm{D})}_{i,k}(u), \chi^{(\mathrm{D})}_{i,k}(u), \mathbf{e}_i\right) \mathrm{d}u + \int_0^s g^{(\mathrm{D})}_\phi\left(\mathbf{h}^{(\mathrm{D})}_{i,k}(u), \chi^{(\mathrm{D})}_{i,k}(u), \mathbf{e}_i\right) \odot \mathrm{d}\mathbf{W}^{(\mathrm{D})}_{i,k}(u), \tag{S6}$$

which has a unique strong solution under standard Lipschitz and linear-growth conditions and is discretized by the Euler–Maruyama update in Eq. (9).

Continuous-time form of Li-GRU (Section III-C, with Eqs. (17)–(20)). During $(t_{k-1}, t_k]$, the gates $\mathbf{z}_{i,k}$, the candidate state $\tilde{\mathbf{h}}_{i,k}$, and the time constant $\boldsymbol{\tau}_{i,k}$ are fixed, and the hidden state follows the first-order relaxation

$$\frac{\mathrm{d}\mathbf{h}_i}{\mathrm{d}t} = \mathbf{z}_{i,k} \odot \left(\tilde{\mathbf{h}}_{i,k} - \mathbf{h}_i\right) \oslash \boldsymbol{\tau}_{i,k}, \qquad \mathbf{h}_i(t_{k-1}) = \mathbf{h}^+_{i,k-1}. \tag{S7}$$

Because the equation is linear and elementwise with constant coefficients, its solution at t_k is

$$\mathbf{h}_i(t_k) = \tilde{\mathbf{h}}_{i,k} + \left(\mathbf{h}^+_{i,k-1} - \tilde{\mathbf{h}}_{i,k}\right) \odot \exp\left[-(t_k - t_{k-1})\, \mathbf{z}_{i,k} \oslash \boldsymbol{\tau}_{i,k}\right] = \boldsymbol{\vartheta}_{i,k} \odot \mathbf{h}^+_{i,k-1} + \left(\mathbf{1}_H - \boldsymbol{\vartheta}_{i,k}\right) \odot \tilde{\mathbf{h}}_{i,k}, \tag{S8}$$

which is Eqs. (19) and (20). A long interval or a small time constant drives $\boldsymbol{\vartheta}_{i,k}$ toward zero and the state toward the candidate, whereas a short interval or a large time constant retains the previous state.

Node feature, history window, and state similarity (Section III-D). The node feature used by Li-GAT, the history window of length $L_{i,k}$ in Eqs. (25) and (26), and the cosine similarity used in Eq. (28) are

$$\mathbf{v}_{i,k} = \left[\mathbf{h}^\top_{i,k}, \bar{x}_{i,k}, \sigma_{i,k}, r_{i,k}, \mathbf{e}^\top_i\right]^\top \in \mathbb{R}^{H+E+3}, \tag{S9}$$

$$\mathcal{W}_{i,k} = \left\{k' \;\middle|\; \max(1, k - L_{i,k} + 1) \leq k' \leq k\right\}, \tag{S10}$$

$$\varsigma_{ij,k} = \frac{\mathbf{v}^\top_{i,k} \bar{\mathbf{v}}^{(i)}_{j,k}}{\|\mathbf{v}_{i,k}\|_2 \, \|\bar{\mathbf{v}}^{(i)}_{j,k}\|_2 + \varepsilon}. \tag{S11}$$

Reconstruction losses (Section III-E, in Eq. (33)). The squared and absolute reconstruction terms are computed over the training targets:

$$\mathcal{L}_{\mathrm{rec}} = \frac{1}{|\Omega^{(\mathrm{tr})}_{\mathrm{tar}}|} \sum_{(i,k) \in \Omega^{(\mathrm{tr})}_{\mathrm{tar}}} \left(\hat{x}_{i,k} - x^\star_{i,k}\right)^2, \qquad \mathcal{L}_{\mathrm{abs}} = \frac{1}{|\Omega^{(\mathrm{tr})}_{\mathrm{tar}}|} \sum_{(i,k) \in \Omega^{(\mathrm{tr})}_{\mathrm{tar}}} \left|\hat{x}_{i,k} - x^\star_{i,k}\right|. \tag{S12}$$